

- Solve all the problems below in **3 hours**.
- All solutions must be correct, complete and well written.
- Scoring: **1** → 4 points, **2,3,4,5,6** → 3.5 points.

1. Answer by true or false with justification.

- (a) (0.75 point) If $f : [a, b] \rightarrow \mathbb{R}$ satisfies $f(a)f(b) \leq 0$, then there is $c \in [a, b]$ such that $f(c) = 0$. False, see for example

$$f(x) = \begin{cases} 4, & \text{for } -1 \leq x < 0, \\ -10, & \text{for } 0 \leq x \leq 3. \end{cases}$$

- (b) (0.75 point) If $f : [a, b] \rightarrow \mathbb{R}$ is monotonic and satisfies $f(a)f(b) \leq 0$, then there is $c \in [a, b]$ such that $f(c) = 0$. False, see for example

$$f(x) = \begin{cases} x - 1, & \text{for } -1 \leq x < 0, \\ x + 1, & \text{for } 0 \leq x \leq 15. \end{cases}$$

- (c) (0.75 point) If $f : \Omega \rightarrow \mathbb{R}$ satisfies $f'(x) = 0$ for all $x \in \Omega$ where $\Omega \subset \mathbb{R}$, then f is constant on Ω . False, see for example

$$f(x) = \begin{cases} -5, & \text{for } -1 < x < 0, \\ 17, & \text{for } 3 < x < 7. \end{cases}$$

- (d) (0.75 point) If $f :]a, b[\rightarrow \mathbb{R}$ is uniformly continuous on $]a, b[$ with $a, b \in \mathbb{R}$, then f is continuous on $[a, b]$. False, the function is not defined at the endpoints, so it can not be continuous.

- (e) (0.5 point) Let f, g be two differentiable functions at 0. If $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{g(x) - g(0)} = \ell$ then $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)} = \ell$. False, see for example

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & \text{for } x \neq 0, \\ 0, & \text{for } x = 0. \end{cases}$$

and

$$g(x) = x$$

- (f) (0.5 point) For any differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, the function $|f|$ is not differentiable at any $a \in \mathbb{R}$ with $f(a) = 0$. False, see for example

$$f(x) = x^3 \text{ at } x = 0$$

2. Let $\omega = \exp\left(i\frac{2\pi}{7}\right)$ and consider the complex numbers

$$A = \omega + \omega^2 + \omega^4 \quad \text{and} \quad B = \omega^3 + \omega^5 + \omega^6.$$

- (a) (2 points) Show that $A = \overline{B}$ and $1 + A + B = 0$.

$$\begin{aligned} \text{We have } \overline{B} &= \overline{\omega^3 + \omega^5 + \omega^6} = \frac{1}{\omega^3} + \frac{1}{\omega^5} + \frac{1}{\omega^6} \quad (\text{since } |w| = 1) \\ &= \frac{\omega^4}{\omega^7} + \frac{\omega^2}{\omega^7} + \frac{\omega^1}{\omega^7} = \omega + \omega^2 + \omega^4 = A, \quad (\omega^7 = 1). \end{aligned}$$

- (b) (1.5 points) Show that

$$\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right) = -\frac{1}{2}.$$

One has

$$1 + A + B = 1 + \omega + \omega^2 + \omega^4 + \omega^3 + \omega^5 + \omega^6 = \frac{1 - \omega^7}{1 - \omega} = 0$$

but

$$1 + A + B = 1 + A + \overline{A} = 0$$

Thus

$$\operatorname{Re}(A) = \frac{-1}{2}.$$

From another hand we have

$$\begin{aligned} \operatorname{Re}(A) &= \operatorname{Re}(\omega) + \operatorname{Re}(\omega^2) + \operatorname{Re}(\omega^4) = \cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{8\pi}{7}\right) \\ &= \cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(2\pi - \frac{6\pi}{7}\right) = \cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right) \end{aligned}$$

this completes the solution.

3. Let $a > 0$, and let $(u_n)_{n \geq 0}$ be a sequence defined by $u_0 > 0$ and

$$u_{n+1} = \frac{1}{2} \left(u_n + \frac{a}{u_n} \right), \quad n \geq 0.$$

- (a) (2 points) Show, for every $n \geq 1$, that $u_n \geq \sqrt{a}$.

First method: By AM-GM, we got

$$u_{n+1} = \frac{1}{2} \left(u_n + \frac{a}{u_n} \right) \geq \sqrt{a}.$$

Second method : By

$$u_{n+1}^2 - a = \frac{(u_n^2 - a)^2}{4u_n^2},$$

for all $n \geq 0$, we have $u_{n+1}^2 - a > 0$ and since u_{n+1} is positive, then $u_{n+1} > \sqrt{a}$.

- (b) (1.5 points) Show that $(u_n)_{n \geq 1}$ is non-increasing, and deduce that (u_n) converges to $\sqrt{a}$. Indeed we have

$$u_{n+1} - u_n = \frac{1}{2} \left(u_n + \frac{a}{u_n} \right) - u_n = \frac{1}{2} \frac{a - u_n^2}{u_n} \leq 0.$$

therefore the sequence is nonincreasing and bounded from below, so it is convergent. The limit satisfies

$$\ell = \frac{1}{2} \left(\ell + \frac{a}{\ell} \right).$$

we keep only $\ell = \sqrt{a}$ as a limit.

4. (3.5 points=1+1+1.5) Find all the pairs $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ for which the function

$$g(x) = \begin{cases} \ln(x+1), & \text{for } x \geq 0, \\ \alpha \sin(x+\beta), & \text{for } x < 0, \end{cases}$$

is continuous at 0.

Clearly we have $g(0) = \ln(1) = 0$ and

$$g(0^+) = \lim_{x \rightarrow 0^+} g(x) = \ln(1) = 0 = g(0).$$

Also, we have

$$g(0^-) = \lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^-} \alpha \sin(x+\beta) = \alpha \sin \beta$$

Since g is continuous at 0 if and only if $g(0^+) = g(0^-) = g(0) = 0$, we conclude that g is continuous at 0 if and only if

$$\alpha \sin \beta = 0 \Rightarrow \begin{cases} \alpha = 0, \\ \text{or} \\ \sin \beta = 0 \end{cases}$$

then

$$(\alpha, \beta) \in \{(0, \gamma) : \gamma \in \mathbb{R}\} \cup \{(\gamma, k\pi) : \gamma \in \mathbb{R} \text{ and } k \in \mathbb{Z}\}.$$

5. Let $a, b \in \mathbb{R}$ with $a < b$. Let $f : [a, b] \rightarrow [a, b]$ be a continuous function.

- (a) (2 points) Show that there exists $c \in [a, b]$ for which $f(c) = c$. Let A be any number between $f(a)$ and $f(b)$. We shall show that there exists a number $c \in]a, b[$ such that $f(c) = c$. Consider a function ϕ defined on $[a, b]$ such that

$$\phi(x) = f(x) - x$$

Clearly $\phi(x)$ is continuous on $[a, b]$.

Also

$$\phi(a) = f(a) - a, \text{ and } \phi(b) = f(b) - b$$

are of opposite signs.

Thus the function ϕ is continuous on $[a, b]$ and $\phi(a)$ and $\phi(b)$ are of opposite signs; therefore, by the Intermediate value theorem, $\exists c \in]a, b[$ such that

$$\begin{aligned}\phi(c) &= 0 \\ f(c) - c &= 0 \Rightarrow f(c) = c.\end{aligned}$$

(b) (1.5 points) Show that if f satisfies

$$|f(x) - f(y)| \leq \frac{1}{2}|x - y|, \quad \forall x, y \in [a, b],$$

then there exists a unique $c \in [a, b]$ for which $f(c) = c$. Suppose by contra positive the existence of two elements such that $f(c_1) = c_1$ and $f(c_2) = c_2$. One has

$$|c_1 - c_2| = |f(c_1) - f(c_2)| \leq \frac{1}{2}|c_1 - c_2|, \quad \forall c_1, c_2 \in]a, b[,$$

thus

$$|c_1 - c_2| \leq \frac{1}{2}|c_1 - c_2| \implies c_1 = c_2.$$

6. Let a be a positive number, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying

$$f(x + a) = \frac{1}{2} + \sqrt{f(x) - f(x)^2}, \quad \forall x \in \mathbb{R}.$$

Show that f is uniformly continuous on $\mathbb{R}$.

Hint. Check the periodicity.

(2 points) First of all, we deduce, without difficulty, from the functional equation that f satisfies

$$\frac{1}{2} \leq f(x) \leq 1, \quad \forall x \in \mathbb{R}.$$

Then, we have

$$\begin{aligned}f(x + a + a) &= \frac{1}{2} + \sqrt{f(x + a) - f(x + a)^2}, \quad \forall x \in \mathbb{R}, \\ &= \frac{1}{2} + \sqrt{\frac{1}{2} + \sqrt{f(x) - f(x)^2} - \left(\frac{1}{2} + \sqrt{f(x) - f(x)^2}\right)^2}, \quad \forall x \in \mathbb{R}, \\ &= \frac{1}{2} + \sqrt{\frac{1}{4} - f(x) + f(x)^2} = \frac{1}{2} + \sqrt{\left(f(x) - \frac{1}{2}\right)^2} = f(x), \quad \forall x \in \mathbb{R}.\end{aligned}$$

which means that f is $2a$ -periodic.

(2 points) Therefore f is continuous and periodic so it is uniformly continuous.